

MICHIGAN DEPARTMENT OF ENVIRONMENT, GREAT LAKES, AND ENERGY

INTEROFFICE COMMUNICATION

TO: Claire Lowe, Cadillac District Office, Water Resources Division

FROM: Phillip DePetro, Permits Section, Water Resources Division

DATE: April 19, 2022

SUBJECT: Local Limits Review (MiWaters Submission HPG-XCC7-NCG09)
Cadillac WWTP
National Pollutant Discharge Elimination System (NPDES) Permit No. MI0020257

Permits Section has evaluated your request for a local limits review for the Cadillac WWTP. This memo summarizes the calculated effluent limits for five-day carbonaceous biochemical oxygen demand (CBOD₅), ammonia (NH₃-N), minimum effluent dissolved oxygen (DO), and total suspended solids (TSS). The Cadillac WWTP discharges treated municipal wastewater to the Clam River at 44.2657°N, -85.3985°E in Wexford County. The facility has an annual average design flow of 3.2 million gallons per day (MGD)

Note that the calculated limits provided in this memo are based on water quality and not the current permit effluent limits. In most cases the effluent limits set in the current permit are based on anti-degradation. In the 1980s, WQBELs were developed based on a trout stream designation (coldwater protection) for the Clam River between Lake Cadillac and the Wexford-Missaukee County line in the non-summer seasons. The Michigan DNR Fisheries order in 1997 and the most current fisheries order list the Clam River as being protected for coldwater fish downstream of the Wexford-Missaukee County line, and as warmwater protected between the lake and the county line. The difference in stream designation between the most recent modeling and the modeling done in the 1980's is the primary driver for the difference in limits.

The Clam River from Lake Cadillac to the Wexford-Missaukee county line is protected for warmwater fish species, other indigenous aquatic life and wildlife, agriculture, navigation, industrial water supply, public water supply at the point of intake, partial body contact recreation, total body contact recreation from May 1 to October 31, and fish consumption. The Clam River downstream of the county line is protected for the same designated uses in addition to being protected for coldwater fish.

The monthly exceedance flows (in cfs) for the Clam River at the Cadillac WWTP outfall are as follows based on Low Flow Discharge Request 10314:

	<u>JAN.</u>	<u>FEB.</u>	<u>MAR.</u>	<u>APR.</u>	<u>MAY</u>	<u>JUNE</u>
50%	17	17	36	78	35	13
95%	5.6	5.1	9.0	14	6.9	4.1
	<u>JULY</u>	<u>AUG.</u>	<u>SEP.</u>	<u>OCT.</u>	<u>NOV.</u>	<u>DEC.</u>
50%	7.4	6.3	6.3	10	18	19
95%	3.0	3.0	3.1	3.7	5.5	5.5

The lowest monthly 95% exceedance flows in up to four seasons are used as the background flow of the receiving stream for the purposes of developing WQBELs. The following seasons are used, consistent with the current permit structure: June-September, October-November, December-March, and April-May.

A multiple reach Streeter-Phelps model was used to model the impact of the Cadillac WWTP on the Clam River at critical low-flow conditions. The Haring Twp WWTP (NPDES Permit No. MI0059076) was included in this model given its proximity downstream. Effluent limit recommendations are determined that are always protective of the applicable DO standards. A diurnal variation of 0.75 mg/l per day between the daily average and daily minimum DO concentration was assumed based on a 2006 field study for the warmwater-protected stream reaches. A diurnal variation of 0.51 mg/l was assumed for the coldwater-protected reaches based on the same study. Kinetic rates were also set based on department procedure defaults for CBOD₅. A 1983 field study determined that nitrification was not occurring in the Clam River due to a lack of substrate for nitrifying bacteria. Site-specific time of passage and reaeration studies completed in 1983 were used to determine stream velocity and reaeration. Stream slopes were updated using LiDAR data. The recommended local limits based on existing stream designations and water quality for the Cadillac WWTP are presented in Table 1.

As previously stated, the effluent limit recommendations of the Cadillac WWTP in the summer season are less restrictive than current permit effluent limits due to the change in stream designation. The current permit effluent limits are based on anti-degradation in the basis of decision memo associated with the last permit reissuance. There is no daily maximum effluent limit recommendation for NH₃-N based on the assumption that nitrification is not occurring in the Clam River.

Effluent limit recommendations based on federal secondary treatment standards (STS) are sufficient to protect the DO standard in the Clam River in the remaining three seasons. A minimum effluent DO concentration of 3.0 mg/l is recommended to protect against acute toxicity in the effluent-receiving water mixing zone. The STS-based effluent limits are less restrictive than current permit effluent limits for the Cadillac WWTP.

The Cadillac WWTP discharge was also evaluated to determine 30-day average NH₃-N effluent limits that are protective of the chronic NH₃-N toxicity criterion. A new chronic toxicity criterion was adopted by the department in 2019 and is being implemented for permits being issued in the current fiscal year. An effluent pH of 7.3 SU was assumed for the Cadillac WWTP based on two years of discharge monitoring data. The recommended effluent limits to protect against chronic toxicity are more restrictive than current permit effluent limits at the Cadillac WWTP in April. The limits for the remaining months at the Cadillac WWTP are equivalent to or less restrictive than current permit effluent limits.

There is no applicable numeric water quality standard for TSS in the Part 4 Water Quality Standards. The narrative standard set in Rule 50 [R323.1050] would apply for TSS.

These effluent limit recommendations are based on water quality standards. We have not addressed treatment practicality or cost effectiveness. Our recommendations do not imply that other considerations, such as the existing permit effluent limits, should not be taken into account when deciding on permit limits.

Table 2. Cadillac WWTP (MI0020257) Local Limits Review Recommendations

Parameter	Months	Concentration (mg/l)	Basis	Rationale
CBOD ₅	June-September	17	Daily Maximum	DO Standard
	October-May	25	7-Day Average	Secondary Treatment
		40	30-Day Average	Secondary Treatment
NH ₃ -N	June-September	0.9	30-Day Average	Chronic Toxicity
	October-November	3.1	30-Day Average	Chronic Toxicity
	December-March	5.4	30-Day Average	Chronic Toxicity
	April-May	3.0	30-Day Average	Chronic Toxicity
D.O.	June-September	6.0	Minimum	DO Standard
	October-May	3.0	Minimum	Acute DO Toxicity

Design Flow = 3.2 MGD